Sahaj Bhadja

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Summary

Software Engineer with embedded-systems projects and 2+ years developing **C/C++, Python, real-time, and resource-conscious software**, combining low-level debugging, automated validation, performance analysis, and cross-functional delivery across hardware-adjacent systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected asynchronous Python services processing 1K+ concurrent jobs at 3x throughput through **state-machine orchestration, bounded retries, fault isolation, structured diagnostics, automated tests, and detailed production performance analysis**.
- Optimized Azure SQL-backed workflows for 500+ users, reducing dashboard latency 71% through **resource profiling, indexed access patterns, concurrency controls, runtime monitoring, technical documentation, and systematic root-cause debugging**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built concurrent Python ingestion services for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **parallel execution and failure recovery**.
- Delivered React and Flask controls supporting 150+ instructors and 10K+ daily requests, contributing **requirements analysis, code reviews, unit testing, production debugging, release validation, and cross-team technical communication**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Developed Decision Transformer pipelines processing 100K+ autonomous-driving trajectories, improving planning 18% through **Python simulation, timing-sensitive sequence modeling, reproducible experiments, rigorous data validation, and quantitative runtime performance benchmarking**.
- Integrated HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% while **debugging complex system behavior, analyzing edge cases, and documenting reproducible research results**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and asynchronous data pipelines using object-oriented patterns, improving response times 30% while reducing technical debt 20% through **modular interfaces, profiling, caching, and failure isolation**.
- Automated 20+ unit and integration suites across Windows and Linux using PyTest, Mockito, Jenkins, and virtual machines, strengthening **CI/CD, regression detection, reproducible validation, and dependable multi-platform releases**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client requirements into technical designs for 12+ integrations, mapping data flows, APIs, synchronization, and failure scenarios while **communicating system behavior clearly** and reducing design errors 25%.
- Built Flask services supporting 50K+ records with sub-200-millisecond responses, applying **unit testing, integration debugging, systematic performance validation, release support, and stakeholder demonstrations** across reliable client-facing production workflows.

Skills

Embedded Development: C, C++, Embedded C, Bare Metal, Microcontrollers, Interrupts, Device Drivers, Bootloaders, RTOS, Memory-Constrained Systems, State Machines
Wireless & Hardware: RF Fundamentals, Bluetooth Low Energy, Zigbee, Thread, Packet Communication, Modulation, GPIO, SPI, I2C, UART, JTAG, SWD, Logic Analyzers, Oscilloscopes
Software & Scripting: Python, Java, Linux, Git, GCC, CMake, Bash, Multithreading, Concurrency, Data Structures, Algorithms, Object-Oriented Programming
Validation & Delivery: Unit Testing, Integration Testing, Hardware-in-the-Loop, Test Automation, CI/CD, Jenkins, GitHub Actions, Debugging, Profiling, Code Reviews, Technical Documentation
Systems: Real-Time Software, Radio Control, Transmit/Receive, Power Management, Timing, Diagnostics, Multiprotocol Scheduling, Embedded Security, Performance Optimization

Projects

Wireless RAIL Controller | *C, C++, ARM Cortex-M, BLE, SPI, JTAG*

- Developed a **portable radio abstraction layer** for transmit, receive, packet timing, state transitions, and low-power modes, using interrupt-driven buffers, register-level peripherals, deterministic callbacks, and diagnostic trace hooks.

RF Hardware-in-the-Loop Harness | *Python, C, CMake, GitHub Actions, UART*

- Built an **automated firmware validation system** exercising packet loss, timing boundaries, sleep transitions, malformed frames, and multiprotocol contention through Python-controlled devices, serial telemetry, reproducible tests, and failure artifacts.

CPU Scheduling Simulator | *C++, React, JavaScript, Priority Queues, Multithreading* | Live

- Implemented performance-sensitive C++ algorithms for FCFS, SJF, SRTF, Round Robin, priority, and multilevel queues, validating correctness and visualizing runtime behavior across **interactive scheduling, resource allocation, and concurrency simulations**.